

AIGR Deep Research Report — PIK3R4 / VPS15 (Q99570)

Hypothesis under review

Human PIK3R4/VPS15 (Q99570) localizes to a nucleus-vacuole junction (GO:0071561).

Distinguish the defined organelle contact site from generic nuclear contact or lysosomal trafficking, and evaluate whether the human VPS34/VPS15 complex participates in a relevant contact-site structure, without treating a predominant endosomal/autophagic location as proof of exclusivity.

- **Focus type:** function_assignment
 - **Term in question:** GO:0071561 — *nucleus-vacuole junction* (cellular_component)
 - **Source review:** `genes/human/PIK3R4/PIK3R4-ai-review.yaml`
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Summary

The nucleus-vacuole junction annotation on human PIK3R4 (GO:0071561) is **not supported by any human experimental evidence**. It is an **IBA-only, phylogenetically propagated (PAINT) annotation** carried over from experimentally annotated *Saccharomyces cerevisiae* orthologs into a taxon that does not possess a vacuole and has no described nucleus-vacuole junction. Every piece of positive, human-specific evidence places the PIK3R4-containing class III PI3K complex at **Rab7-positive late endosomes and autophagosomes**, not at the nuclear envelope or a defined nuclear–lysosomal contact site.

Three independent lines of reasoning converge on this conclusion. First, the term is structurally organism-specific: GO:0071561 is explicitly defined as the contact formed between the **vacuole membrane and the outer nuclear membrane**, mediated in *S. cerevisiae* by the direct interaction of **Vac8p and Nvj1p** — two proteins that build the junction and are unrelated to VPS15/VPS34. Second, the human annotation has no experimental basis: the only

GO:0071561 annotation on Q99570 is evidence code IBA (ECO:0000318, GO_REF:0000033), inferred from the yeast ortholog via PANTHER node PTN000426471. Third, the annotation is internally inconsistent — only PIK3R4 carries the human NVJ term while its obligate catalytic partner PIK3C3/VPS34 (Q8NEB9) does not.

Verdict: over-annotated / refuted for human. The most important caveat is that absence of a curated human NVJ annotation is not absolute proof that human class III PI3K never visits a nuclear-envelope contact site — human nuclear-lysosomal / nuclear-envelope membrane-contact biology is an active field and remains incompletely mapped. However, the specific GO:0071561 term denotes a defined yeast organelle, and there is currently **zero positive human evidence** for it. Curation should reflect the state of evidence, not a phylogenetic guess.

Executive Judgment

Verdict: Over-annotated / refuted for human.

Three converging lines of evidence:

1. **The term is structurally organism-specific.** GO:0071561 is the contact between the vacuole membrane and the outer nuclear membrane, built in yeast by Vac8p and Nvj1p. It is not architecturally a "Vps15 structure."
 2. **The human annotation has no experimental basis.** The only GO:0071561 annotation on Q99570 is IBA, inferred from the yeast ortholog via PANTHER PTN000426471. All human *experimental* CC annotations are endosomal/autophagosomal.
 3. **The annotation is internally inconsistent.** Only PIK3R4 carries the human NVJ term; PIK3C3/VPS34 does not. A genuine complex-resident localization would annotate both subunits. The single-subunit, IBA-only pattern is the signature of an isolated PAINT propagation artifact.
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Evidence Matrix

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence & limitations
PMID: 28533415	Structural (crystal)	Qualifies (defines the term)	What builds the NVJ	NVJ is mediated by direct Vac8p–Nvj1p interaction; junction defined by these two proteins	<i>S. cerevisiae</i>	High. Established organism-specific architecture not Vps15 based
PMID: 42375028	Review	Qualifies	Nature of NVJ	NVJ is "a central membrane contact site in yeast" connecting nuclear ER and vacuole	Yeast (review)	High as orientation review-level
PMID: 23335340	Localization (IDA)	Supports (in yeast only)	Do Vps15/ Vps34 sit at NVJ?	Yeast VPS15 (P22219) and VPS34 (P22543) carry IDA NVJ annotations	<i>S. cerevisiae</i>	High for yeast; source of the ancestral PANTHER assignment
PMID: 14617358	Localization (IDA)	Refutes (for human)	Where does human PIK3R4 localize?	hVPS34/p150(PIK3R4) complex colocalizes with Rab7 on late endosomes	Human cells	High. Direct human experimental evidence, non-nucle
QuickGO annotation set (Q99570)	Database	Refutes	Evidence basis of human NVJ term	Only NVJ annotation is IBA (GO_REF:0000033), withFrom PANTHER:PTN000426471 + SGD	Human	High. No human experimental support
QuickGO (Q8NEB9)	Database	Refutes	Is the partner subunit annotated?	Human PIK3C3/VPS34 has 0 NVJ annotations	Human	High. Complex-level inconsistency
UniProt Q99570	Database	Refutes	Curated human	Late endosome; cytoplasmic vesicle/	Human	High

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence & limitations
			subcellular location	autophagosome; membrane. No nuclear/ vacuole terms		
NCBI PubMed searches	Computational (literature)	Refutes	Any human NVJ/ nuclear- envelope report?	"PIK3R4 nuclear envelope" = 0; "nucleus lysosome contact site VPS34" = 0	Human	Moderate- high; negative literature search

Key Findings

F001 — GO:0071561 is a structurally yeast-defined organelle contact site

The QuickGO record for GO:0071561 (aspect cellular_component) defines the term as "an organelle membrane contact site formed between the **vacuole membrane and the outer nuclear membrane**. In *S. cerevisiae* these contacts are mediated through direct physical interaction between **Vac8p and Nvj1p**." The synonyms are "nucleus-vacuole membrane contact site" and "NVJ."

The crystal structure of the Vac8p–Nvj1p interface ([PMID: 28533415](#)) confirms that the junction is physically built by these two proteins — "*Formation of the nucleus-vacuole junction (NVJ) is mediated by direct interaction between the vacuolar protein Vac8p and the outer nuclear endoplasmic reticulum membrane protein Nvj1p.*" Disrupting this interaction abolishes NVJs and piecemeal microautophagy of the nucleus (PMN). The "at a glance" review ([PMID: 42375028](#)) reinforces that the NVJ is "*a central membrane contact site in yeast that connects the nuclear endoplasmic reticulum and the vacuole.*"

Implication: The junction is architecturally defined by Vac8p/Nvj1p, not by Vps15/Vps34. Even in yeast, Vps15/Vps34 are *residents* at the junction, not its structural scaffold. The term therefore carries strong organism-specific and organelle-specific meaning that does not transfer cleanly to a vacuole-less taxon.

F002 — The human PIK3R4 NVJ annotation is IBA-only; human localization is endosomal/autophagosomal

In the QuickGO annotation set for UniProtKB:Q99570 (human PIK3R4), the **only** nucleus-vacuole junction annotation is **IBA** (ECO:0000318, GO_REF:0000033 / PAINT), with `withFrom` = **PANTHER:PTN000426471** and **SGD:S000000301** — i.e., inferred phylogenetically from the yeast ortholog, with **no human experimental support**.

By contrast, every human *experimental* CC annotation is non-nuclear:

- **Late endosome** — IDA (ECO:0000314), [PMID: 14617358](#)
- **Membrane** — HDA,

[P 19946888](#)

- **Autophagosome and cytosol** — Reactome TAS

UniProt Q99570 lists subcellular location = **Late endosome; Cytoplasmic vesicle / autophagosome; Membrane** (keywords: Endosome, Cytoplasmic vesicle; no nuclear or vacuolar terms). The primary human study ([PMID: 14617358](#)) reports that "*the hVPS34/p150 complex colocalized with rab7 on late endosomes and hVPS34 activity was dependent on nucleotide cycling of rab7.*" Here p150 is PIK3R4.

Across all organisms, GO:0071561 has ~5,093 gene-product annotations, overwhelmingly **IEA** (GO_REF:0000118) and **IBA** (GO_REF:0000033), spanning plants, nematodes, birds, and fungi — a classic signature of automated / phylogenetic over-propagation of a yeast-specific structural term.

F003 — Yeast Vps15/Vps34 have genuine NVJ localization; the human paralog does not (lineage-specific architecture)

In *S. cerevisiae* (taxon 559292), both **VPS15 (P22219)** and **VPS34 (P22543)** carry **IDA (ECO:0000314)** NVJ annotations from [PMID: 23335340](#), alongside the canonical NVJ builders **NVJ1 (P38881)** and **VAC8 (P39968)** (IDA,).

[P 10888680](#)

So the ancestral source that propagated to PANTHER node PTN000426471 is **experimentally grounded in yeast**.

The crucial asymmetry: the equivalent human evidence is **absent**. No human PIK3R4/VPS34 dataset places the complex at the nuclear envelope or a defined nucleus–lysosome junction. This is exactly the pattern the seed hypothesis asked us to distinguish — *positive evidence of lineage-specific architecture* versus a *yeast-only assumption based on the word "vacuole."* The evidence favors the former: the junction is a real yeast structure that simply has no demonstrated human counterpart for this complex.

F004 — No human literature or database record places PIK3R4 at the nuclear envelope, vacuole, or a nucleus–lysosome junction

Targeted NCBI PubMed searches returned:

- PIK3R4 nuclear envelope → **0 hits**
- nucleus lysosome contact site VPS34 → **0 hits**
- PIK3R4 nucleus → **1 hit**

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P 33906557
 an autophagy/intestinal-barrier review — not a localization study)

The three human PIK3R4 "localization" hits concern autophagy initiation via ULK1

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P 37992308
),

a VPS15/PIK3R4 ciliopathy affecting IFT20 release from the cis-Golgi

(
P 27882921
),

and Vps34/PIK3C3 complex-subunit characterization

(
P 27630019
)

— **all non-nuclear, non-vacuolar**. Combined with the curated UniProt/QuickGO human locations (late endosome, autophagosome, membrane), there is **no positive human evidence** for the seed hypothesis.

F005 — The human NVJ annotation is internally inconsistent (single-subunit, IBA-only)

QuickGO shows that human **PIK3C3/VPS34 (Q8NEB9)** has **0 annotations** to GO:0071561, whereas in yeast **both** Vps15 (P22219) and Vps34 (P22543) carry experimental IDA NVJ annotations

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P 23335340
)

Across all human proteins (taxon 9606), the **only** GO:0071561 annotation is PIK3R4 (IBA, GO_REF:0000033); **no human protein** has any experimental (IDA/IPI/IMP/HDA) NVJ annotation.

A genuine complex-resident localization would annotate **both obligate subunits**. The single-subunit, IBA-only pattern is diagnostic of an **isolated phylogenetic (PAINT) propagation** rather than a real human cellular structure.

Mechanistic Model / Interpretation

The class III PI3K (VPS34/PIK3C3 catalytic + VPS15/PIK3R4 regulatory pseudokinase scaffold) is deeply conserved, and its *core biochemistry* (PI3P production for endosomal sorting and autophagy) transfers across eukaryotes. What does **not** transfer is the specific yeast **nucleus-vacuole junction** organelle, which is a lineage-specific architecture built by Vac8p–Nvj1p.

YEAST (<i>S. cerevisiae</i>)	HUMAN (<i>H. sapiens</i>)
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Nucleus (outer NE)	Nucleus
Nvj1p <-- junction scaffold	(no Nvj1p / Vac8p orthologous NVJ)
Vac8p <-- junction scaffold	
Vacuole membrane	Lysosome (no defined NVJ described for class III PI3K)
+-- Vps15/Vps34 RESIDE at NVJ (IDA,	Vps15(PIK3R4)/Vps34(PIK3C3)
<a "="" 140000000="" 2"="" href="https://pubmed.ncbi.nlm.nih.gov/23335340/" https:="" pubmed.ncbi.nlm.nih.gov="" rel="noopener noreferrer" title="Autophagosome, cytosol, membrane (experimental)</td> </tr> <tr> <td></td> <td> +-- Autophagosome, cytosol, membrane (experimental)</td> </tr> <tr> <td colspan=">PANTHER node PTN000426471 (Vps15 family)	
IBA / PAINT propagation	
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Human PIK3R4 gets GO:0071561 -- WITHOUT any human experimental support, and WITHOUT the partner PIK3C3 receiving the same term.	

The annotation flow is transparent: yeast Vps15/Vps34 were experimentally seen at the NVJ → this became the ancestral state at PANTHER node PTN000426471 → PAINT/IBA propagated the CC term GO:0071561 down to human PIK3R4. The propagation crossed an organelle boundary

that does not exist in human (no vacuole; no demonstrated NVJ), and it landed on only one of the two complex subunits. Both the **biology** (organism-specific organelle) and the **annotation metadata** (IBA-only, single-subunit, cross-taxon frequency bias) point to over-annotation.

Evidence Base

Paper	PMID	Role in this analysis
<i>Mechanistic insight into the nucleus-vacuole junction based on the Vac8p-Nvj1p crystal structure.</i>	28533415	Defines the NVJ as a Vac8p–Nvj1p structure; establishes the term's organism-specific architecture
<i>The nucleus-vacuole junction at a glance.</i>	42375028	Review confirming NVJ is a yeast contact site (nuclear ER ↔ vacuole)
Yeast Vps15/Vps34 NVJ IDA source	23335340	Experimental basis for the ancestral (yeast) NVJ annotation propagated to PANTHER PTN000426471
hVPS34/p150(PIK3R4)–Rab7 late-endosome study	14617358	Direct human experimental localization: late endosome, not NVJ

Database orientation used (labeled as database-level support): QuickGO term and annotation records for GO:0071561, Q99570 (PIK3R4), Q8NEB9 (PIK3C3); UniProt Q99570 subcellular location; PANTHER node PTN000426471.

GO Curation Implications

Lead (requires curator verification): Remove or NOT-qualify GO:0071561 on human PIK3R4 (Q99570).

- The term is a **CC** term denoting a yeast-specific organelle contact site.
- The current human annotation is **IBA-only** (GO_REF:0000033, PANTHER PTN000426471) with **no human experimental support** and **no partner-subunit corroboration** (PIK3C3 lacks it).

- Recommended action: **remove** the human GO:0071561 annotation, or apply a **NOT** qualifier if the curator prefers to preserve the phylogenetic trail while signaling absence of human evidence. As PAINT annotations are generated by the PAINT curation team, the cleaner route may be a **PAINT curation note** at node PTN000426471 flagging that the NVJ CC term should not be propagated beyond fungi (organism-specific organelle), since human/metazoa lack the vacuole and a described NVJ.
- **Retain** the well-supported human CC terms: **late endosome** (GO:0005770, IDA,

P 14617358

), **autophagosome** (GO:0005776), **cytosol**, **membrane**, and **phosphatidylinositol 3-kinase complex, class III** (GO:0005942) / its autophagy-specific variants.
- The **yeast** NVJ annotations on Vps15/Vps34 (P22219/P22543, IDA,

P 23335340

) remain valid and should not be touched.

This avoids a "protein binding" fallback — the informative replacement is the experimentally supported endosomal/autophagosomal localization already present.

Mechanistic Scope

The claim being tested is strictly a **cellular-component / localization** assertion: does the gene product physically reside at a defined nucleus-vacuole junction? This is distinct from PIK3R4's **molecular function** (regulatory pseudokinase scaffold of the class III PI3K / VPS34 complex) and its **biological processes** (autophagy initiation, endosomal PI3P production, vesicular trafficking), all of which are well supported and unaffected by this review.

The refutation applies only to the **direct localization** at GO:0071561. It does not challenge:

- PIK3R4's role as the VPS15 regulatory subunit of class III PI3K.
- Its established late-endosomal, autophagosomal, and membrane localization.
- Any downstream autophagy or trafficking phenotype.

We are separating **direct gene-product localization** (the term under review) from **pathway function** (intact) and from **loss-of-function phenotypes** (irrelevant to this CC assertion).

Conflicts and Alternatives

1. **Paralog/ortholog carry-over (primary alternative, and the most likely explanation):** The human term derives from yeast orthologs via PANTHER PTN000426471. This is not paralog confusion within human, but **cross-species phylogenetic carry-over** of an organism-specific organelle term. The ~5,093-annotation, IEA/IBA-dominated spread of GO:0071561 across plants, nematodes, and birds confirms systematic over-propagation.
 2. **Organism-specific organelle absence:** Humans have no vacuole; the lysosome is the functional analog, but no nucleus-**lysosome** junction resembling the yeast NVJ has been described for the class III PI3K complex. Applying a vacuole-defined term to human cells is a category mismatch unless a human structure is demonstrated.
 3. **Genuine-but-undiscovered human contact site (the seed's steelman):** Nuclear envelope–lysosome and nuclear-envelope membrane contact biology is an active frontier. It remains *possible* that human class III PI3K transiently visits a nuclear-envelope contact site. However, (a) there is currently no positive human data, and (b) even if found, it would likely warrant a **new/more generic** contact-site term rather than the yeast-specific GO:0071561. This keeps the door open scientifically while still supporting removal of the specific term now.
 4. **Single-subunit inconsistency as artifact signal:** The fact that only PIK3R4 (not PIK3C3) carries the term is best explained by annotation mechanics (which ortholog groups were PAINT-annotated), not by a biological scenario in which only the regulatory subunit visits the junction without its catalytic partner.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Human nuclear-envelope contact biology is incompletely mapped	PubMed negative searches; UniProt/QuickGO curated locations	A future human MCS finding could re-open a (generic) contact-site term	Proximity-labeling (APEX/BioID) of PIK3R4 at the nuclear envelope; high-resolution imaging
P 14617358 read at abstract/snippet level only	Snippet verified via QuickGO withFrom + abstract	Confirms late-endosome, but full text may add nuance	Full-text review of localization panels
PANTHER node PTN000426471 propagation logic not directly inspected in PAINT UI	Inferred from QuickGO withFrom fields	Curators may prefer to fix at the PAINT node rather than per-gene	Inspect PTN000426471 in the PAINT/PANTHER tree and add a curation note
Did not exhaustively survey all metazoan orthologs	Spot-checked cross-taxon annotation counts	Confirms breadth of over-propagation but not every case	Systematic audit of GO:0071561 IBA annotations outside fungi
Yeast IDA source (P 23335340) reviewed via database metadata, not full text	QuickGO IDA records	Establishes ancestral validity; not central to human verdict	Full-text confirmation of yeast Vps15/Vps34 NVJ imaging

Discriminating Tests

- Proximity labeling of endogenous PIK3R4** (BioID/TurboID or APEX2) with nuclear-envelope, lysosomal, and endosomal reference markers. If PIK3R4 proximity partners are dominated by Rab7/late-endosome/autophagosome proteins and lack nuclear-envelope/INM proteins, this refutes a human NVJ localization.

2. **Two-subunit co-localization requirement:** Simultaneous imaging of PIK3R4 and PIK3C3. A real complex-resident junction localization must show **both** subunits at the site; PIK3R4-only signal argues against it.
3. **PANTHER/PAINT node audit at PTN000426471:** Confirm the propagation path and evaluate whether GO:0071561 should be restricted to fungal descendants.
4. **Ortholog conservation check for the junction scaffold:** Test whether human has functional orthologs of Nvj1p/Vac8p forming an NVJ-like structure. Absence of the scaffold strengthens the "no human NVJ" conclusion.
5. **Systematic GO:0071561 annotation audit** across non-fungal taxa to quantify the over-propagation and inform a term-scope fix.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Proposed action change - Remove (or NOT-qualify) **GO:0071561 (nucleus-vacuole junction)** on human PIK3R4 (Q99570); prefer fixing at PAINT node **PTN000426471** so the term is not propagated beyond fungi. - Retain: **GO:0005770 late endosome** (IDA,

P 14617358

GO:0005776 autophagosome, GO:0005942 class III PI3K complex, cytosol/membrane. - Leave yeast Vps15/Vps34 (P22219/P22543) NVJ IDA annotations unchanged.

Candidate references with snippets to verify - **PMID: 28533415**: "*Formation of the nucleus-vacuole junction (NVJ) is mediated by direct interaction between the vacuolar protein Vac8p and the outer nuclear endoplasmic reticulum membrane protein Nvj1p.*" → NVJ is a Vac8p/Nvj1p structure, not a Vps15 structure. - **PMID: 42375028**: "*The nucleus-vacuole junction (NVJ) is a central membrane contact site in yeast...*" → yeast-specific organelle. - **PMID: 14617358**: "*The hVPS34/p150 complex colocalized with rab7 on late endosomes...*" → human localization is endosomal.

Suggested curator questions - Should organism-specific CC terms like GO:0071561 be blocked from IBA propagation into taxa lacking the organelle (vacuole)? - Is a NOT qualifier or outright removal preferred for IBA-only, single-subunit, cross-taxon over-propagations?

Suggested experiments — see Discriminating Tests above (PIK3R4 proximity labeling; two-subunit co-localization; PAINT node audit).

Conclusion

The seed hypothesis is **refuted for human as stated**. Human PIK3R4/VPS15 (Q99570) has **no direct evidence** of localizing to a nucleus-vacuole junction. GO:0071561 is a **structurally yeast-defined** contact site built by Vac8p-Nvj1p ([PMID: 28533415](#)); the human annotation is **IBA-only**, phylogenetically propagated (PANTHER PTN000426471 / PAINT) from experimentally annotated yeast Vps15/Vps34 ([PMID: 23335340](#)) into a taxon that lacks a vacuole and any described NVJ. Human PIK3R4 is instead experimentally localized to **Rab7+ late endosomes and autophagosomes** ([PMID: 14617358](#)), and only PIK3R4 — not its obligate partner PIK3C3 — carries the term, marking it an isolated propagation artifact. **Curation lead:** remove or NOT-qualify GO:0071561 on human PIK3R4 while retaining its late-endosome, autophagosome, and class III PI3K complex CC terms; yeast NVJ annotations remain valid.

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